

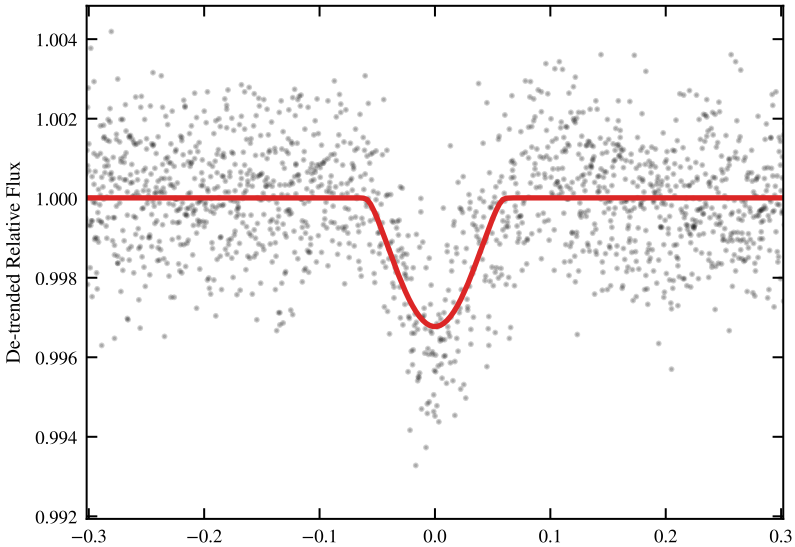
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 261369656 | RA: 292.60481° | Dec: -46.27807° | Sectors: 104, 105

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	10.959 ± 0.000921
Rp/R★	0.2485 ± 0.0993
Rp (R⊕)	56.115 ± 22.7
b	1.1823 ± 0.107
T14 (h)	2.8978 ± 0.144
a (AU)	0.11238 ± 0.00752
Teq (K)	1334.3 ± 39.9

Primary Phased Transit



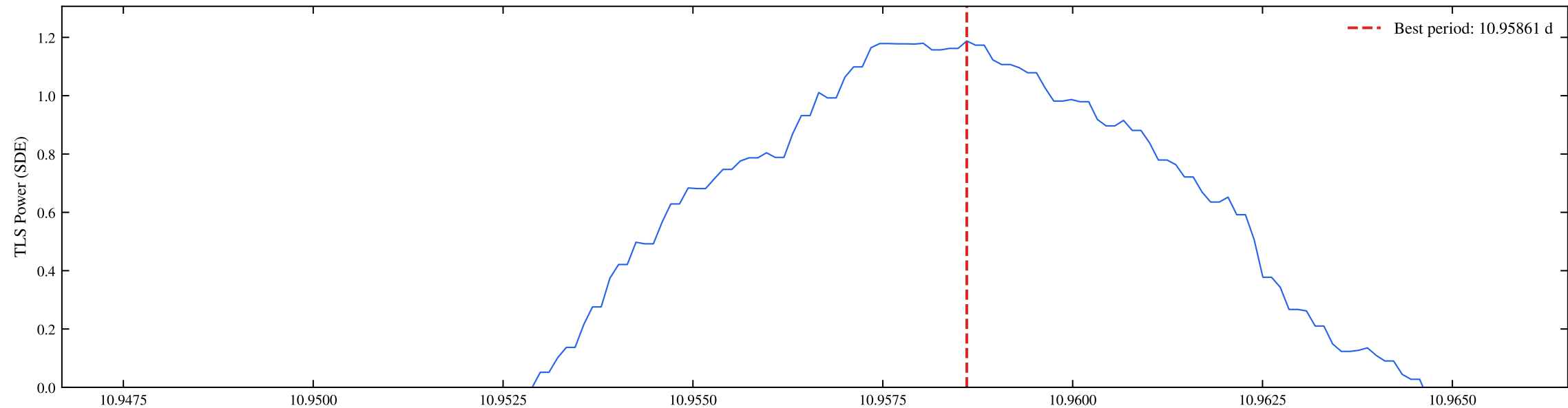
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	10.9586	10.958577 ± 0.00092
P1 Epoch (BTJD)	N/A	4187.1411 ± 0.0022
Max R-hat (R)	N/A	1.0019
Min ESS	N/A	4889
MCMC Divergences	N/A	0

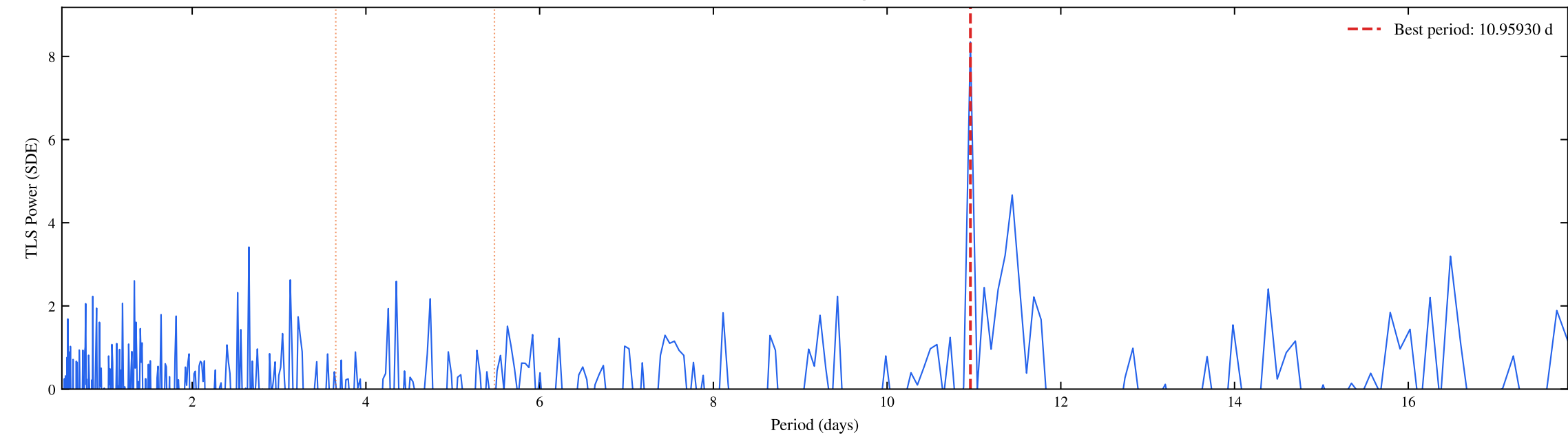
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	2.068 ± 0.103	VizieR TIC8
M★ (M☉)	1.31 ± 0.18	VizieR TIC8
Teff (K)	6456 ± 131	VizieR TIC8
log g (dex)	3.924 ± 0.108	VizieR TIC8
[Fe/H]	-0.5866 ± 0.0118	Gaia DR3
ρ★ (g/cm³)	0.2088 ± 0.0424	Derived

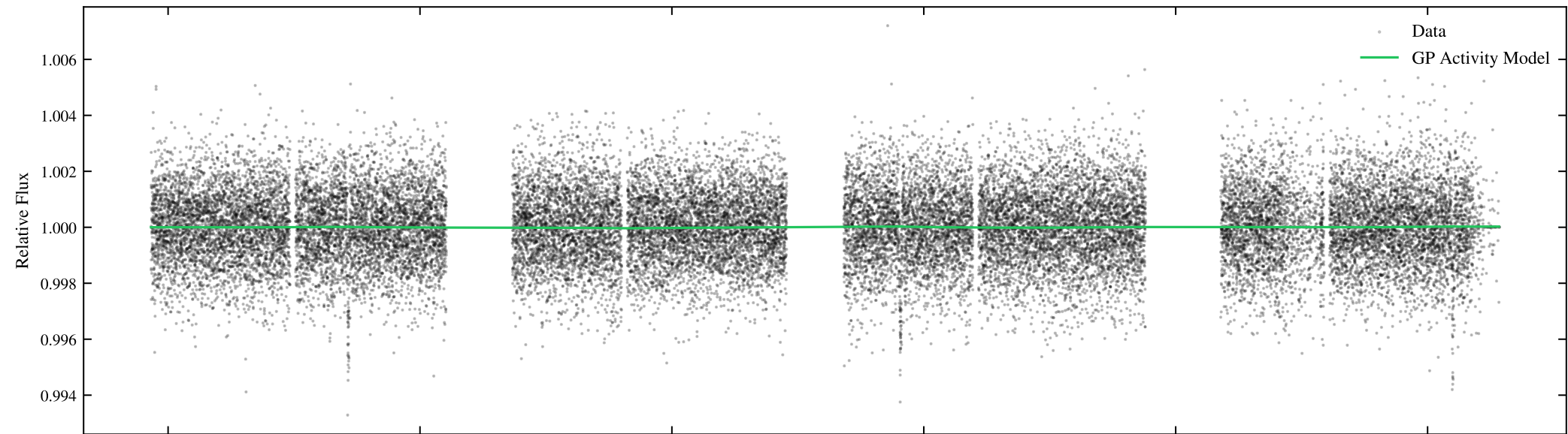
TIC 261369656 | Sectors: 104, 105 | TLS Periodogram (Planet 1 Search) (SDE = 1.19)



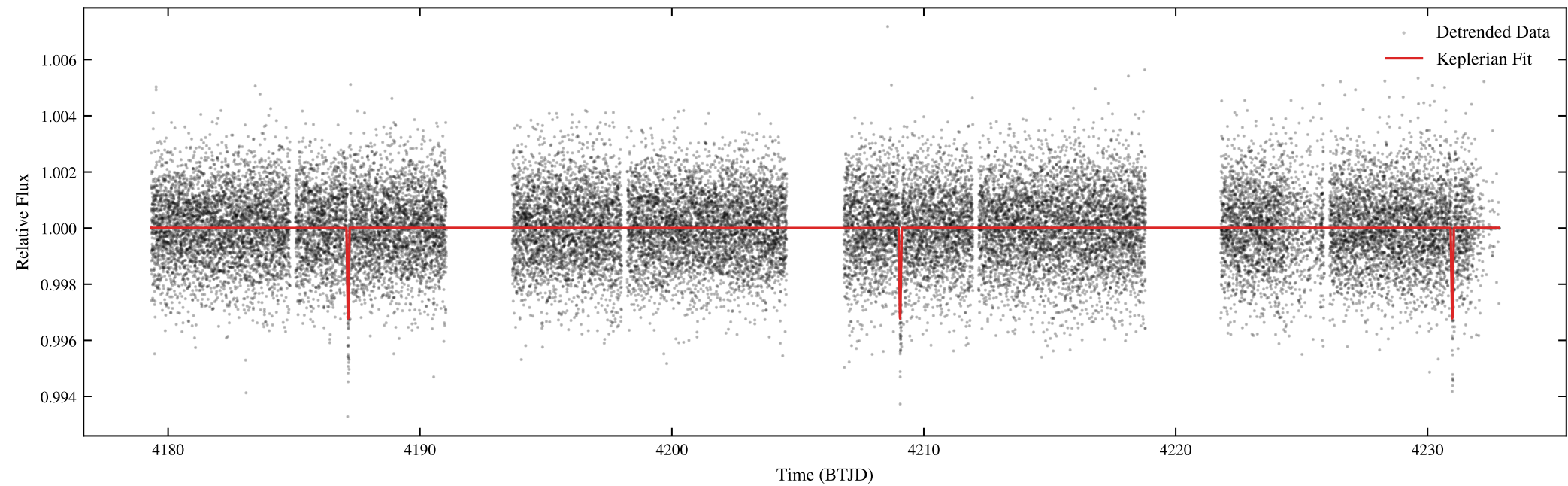
TIC 261369656 | Sectors: 104, 105 | TLS Periodogram (Planet 2 Search) (SDE = 8.34)



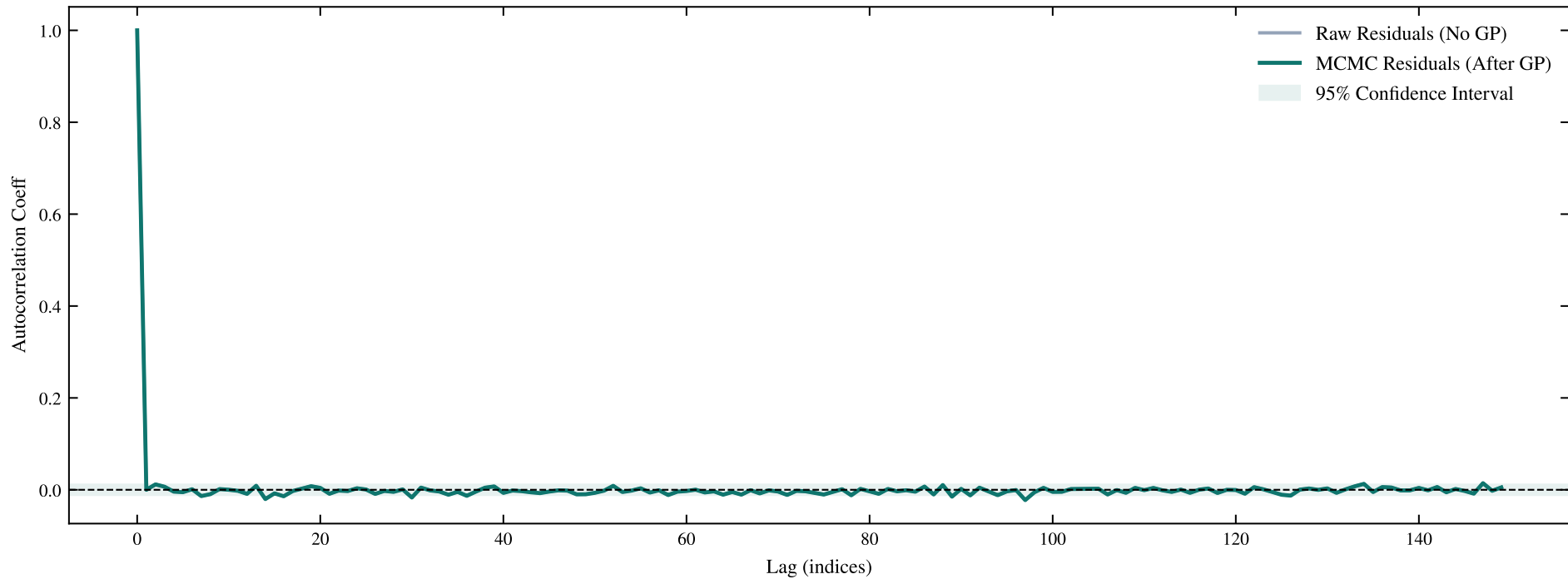
TIC 261369656 | Sectors: 104, 105 | Bayesian GP Detrending (Time Domain)



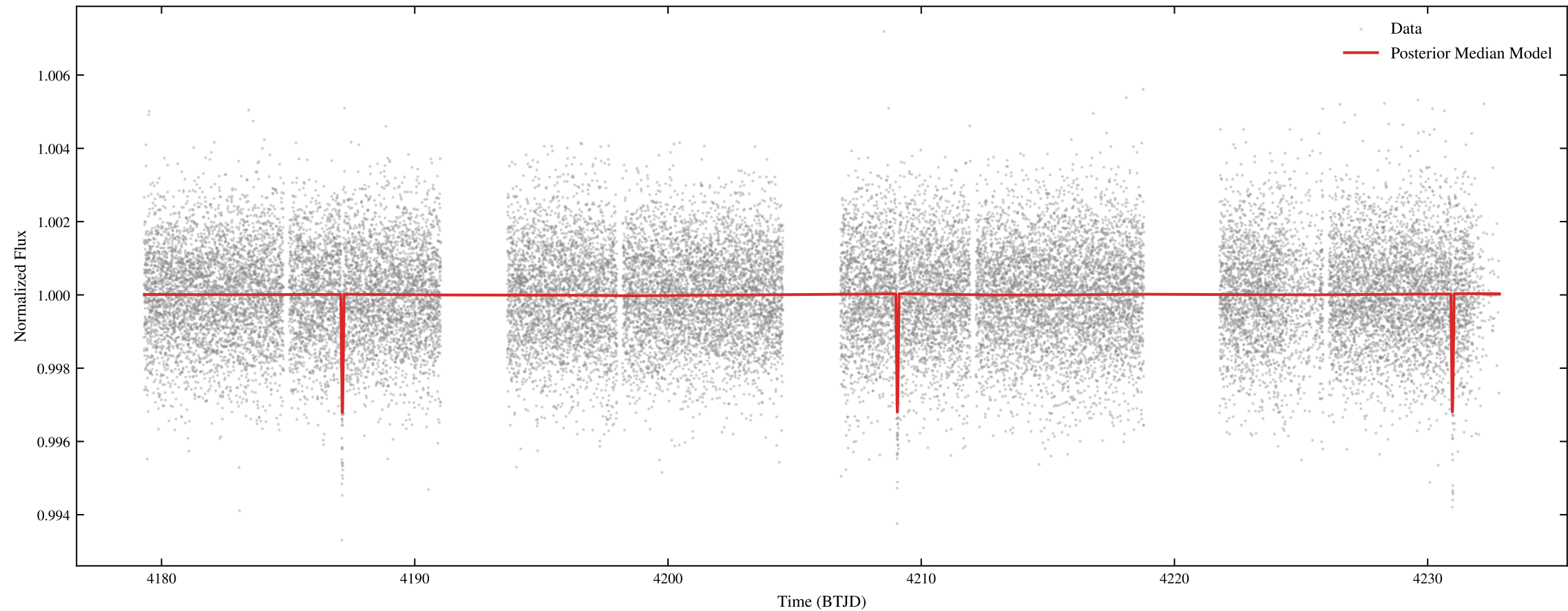
Detrended Light Curve & Best-Fit Transit Model

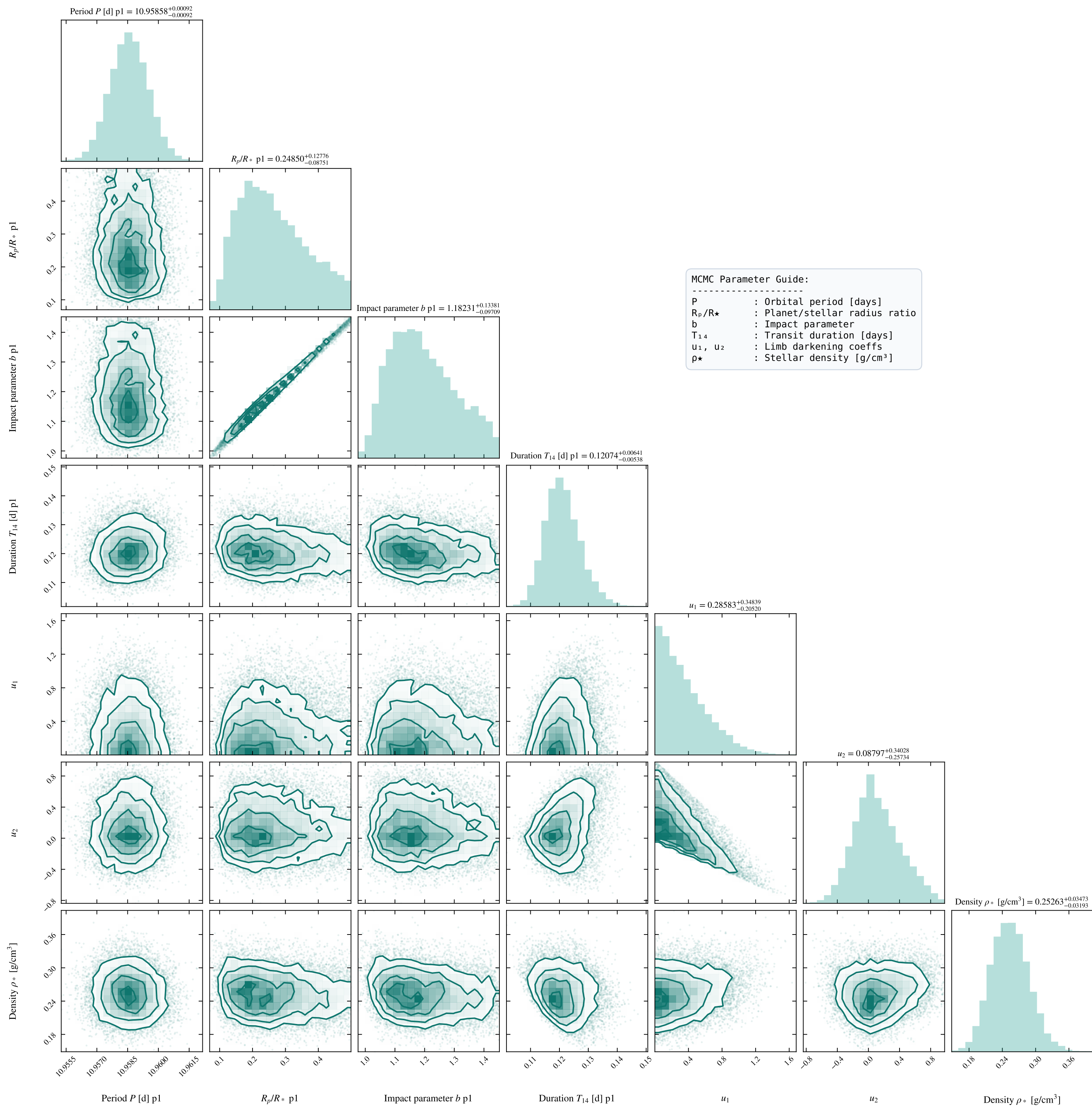


TIC 261369656 | Sectors: 104, 105 | GP Residuals Autocorrelation Function (ACF)

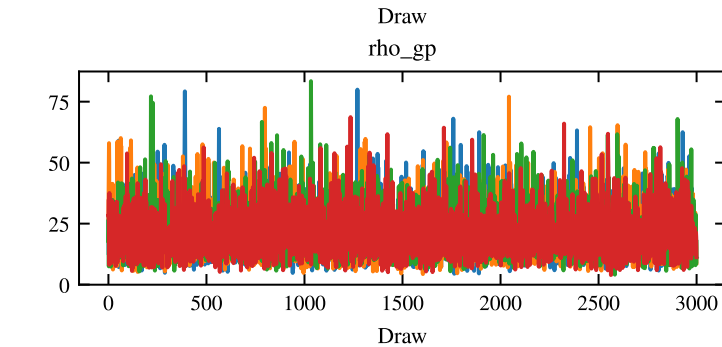
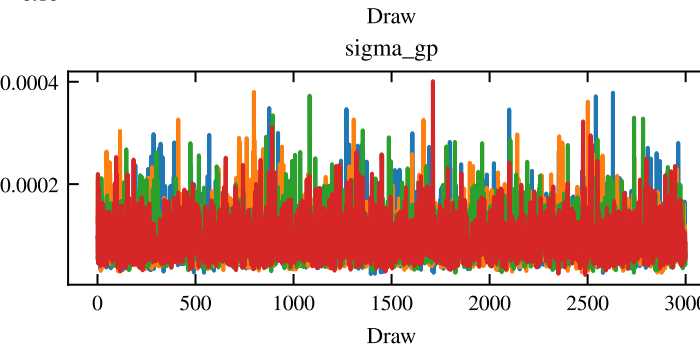
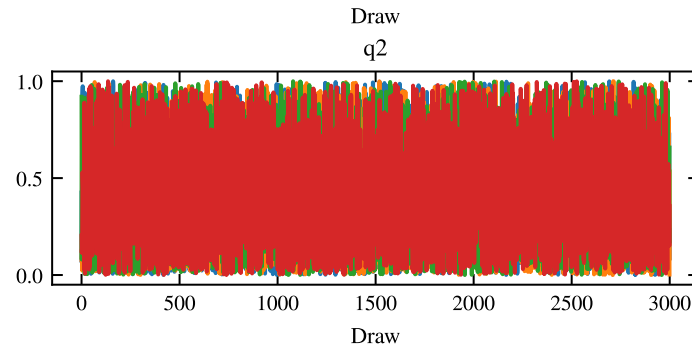
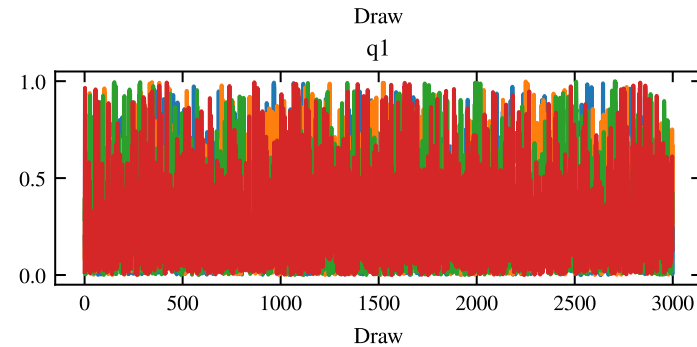
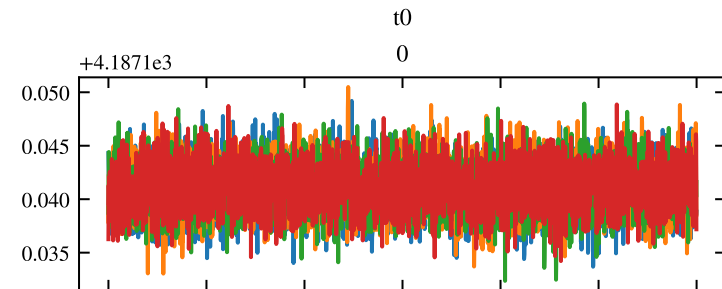
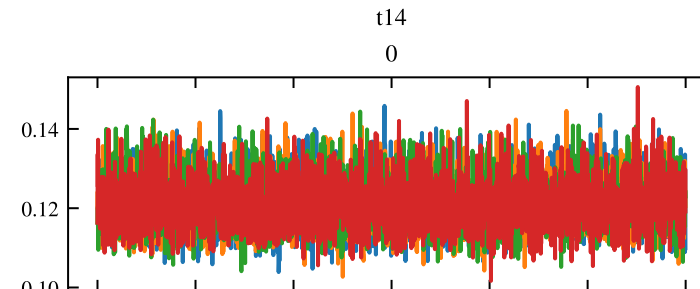
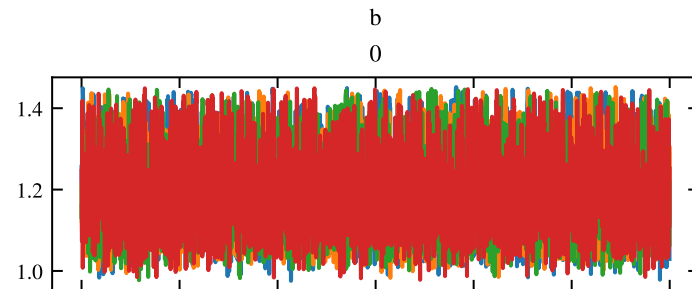
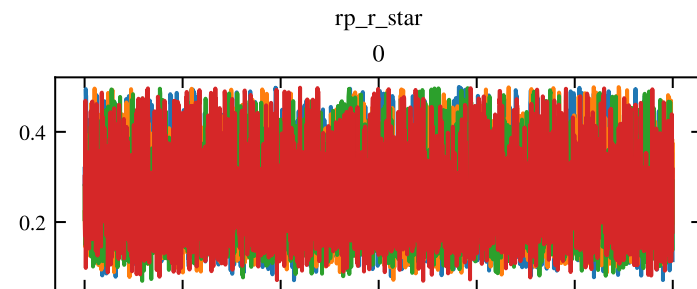


TIC 261369656 | Sectors: 104, 105 | MCMC Posterior Predictive Check

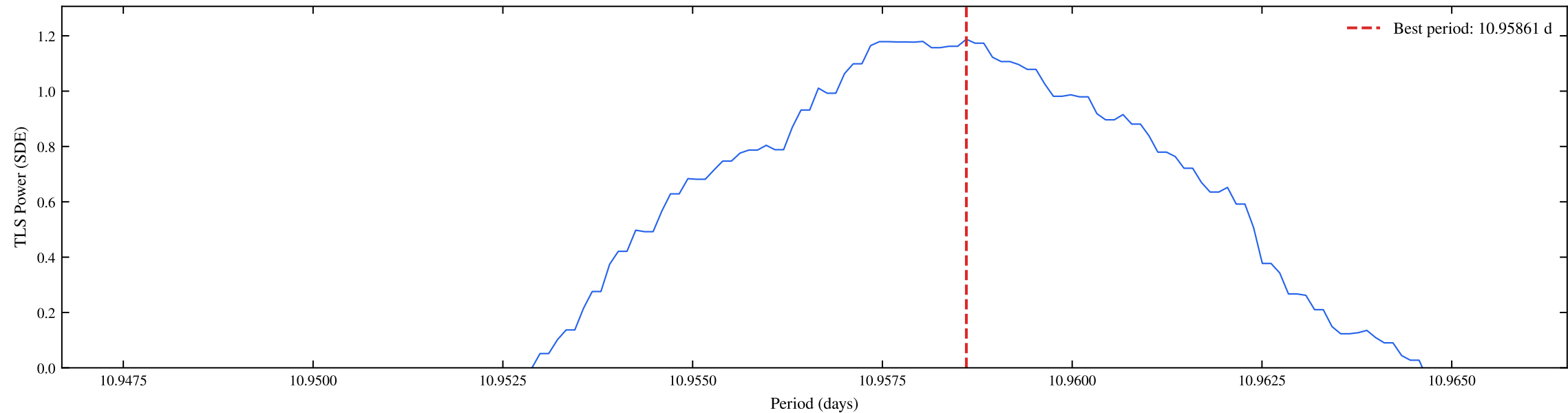




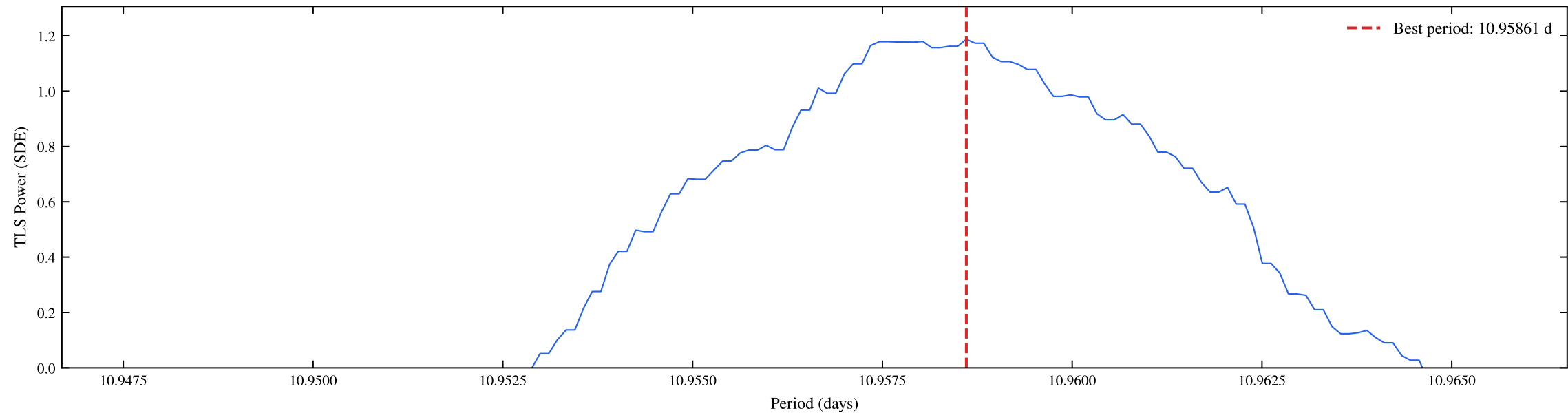
TIC 261369656 | Sectors: 104, 105 | MCMC Trace Plots



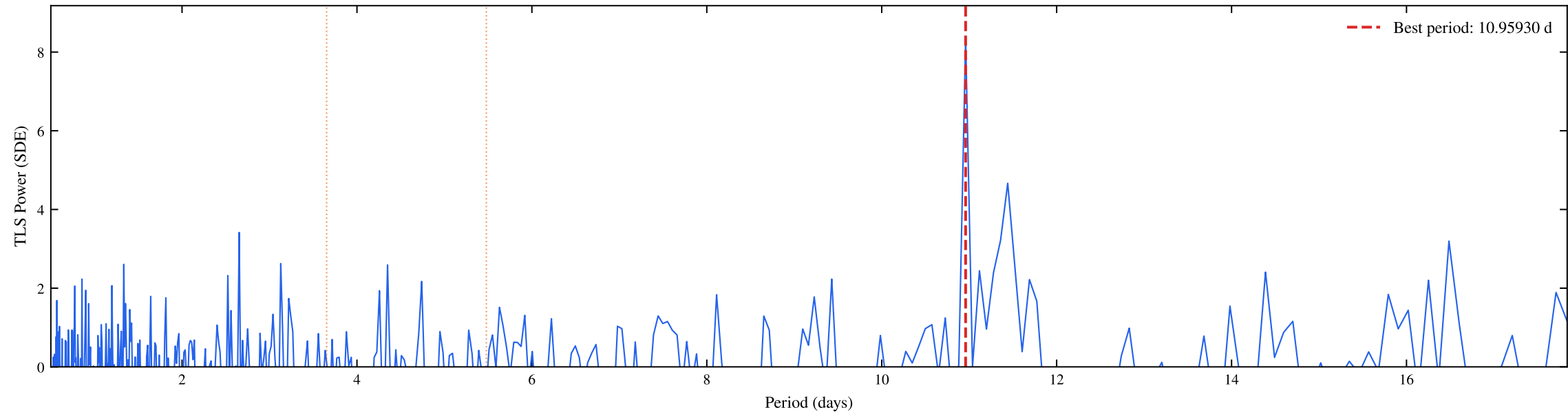
TIC 261369656 | Sectors: 104, 105 | TLS Coarse Periodogram (SDE = 1.19)



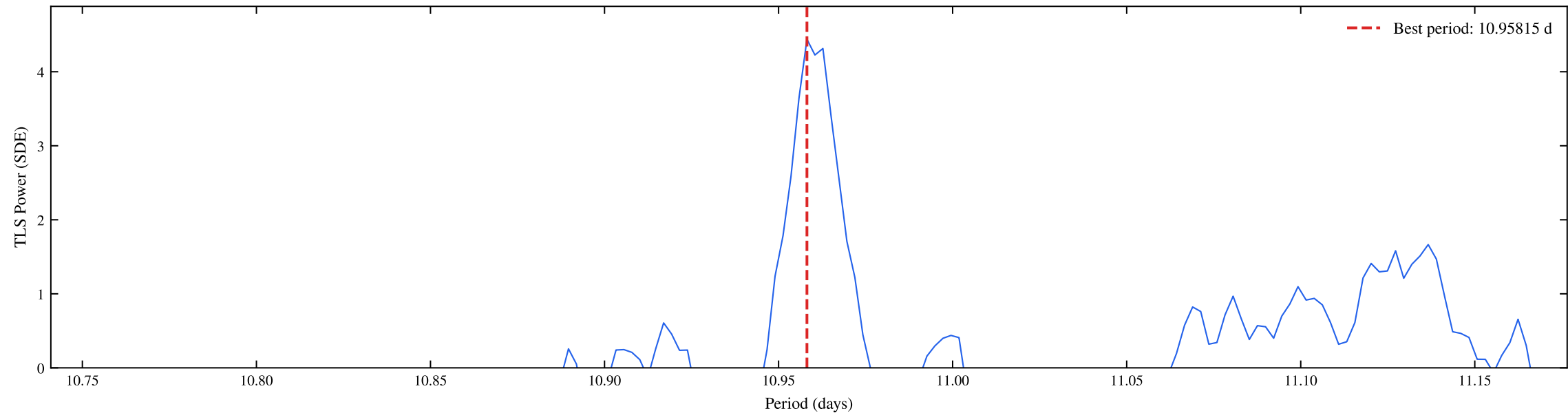
TIC 261369656 | Sectors: 104, 105 | TLS Fine Periodogram (SDE = 1.19)



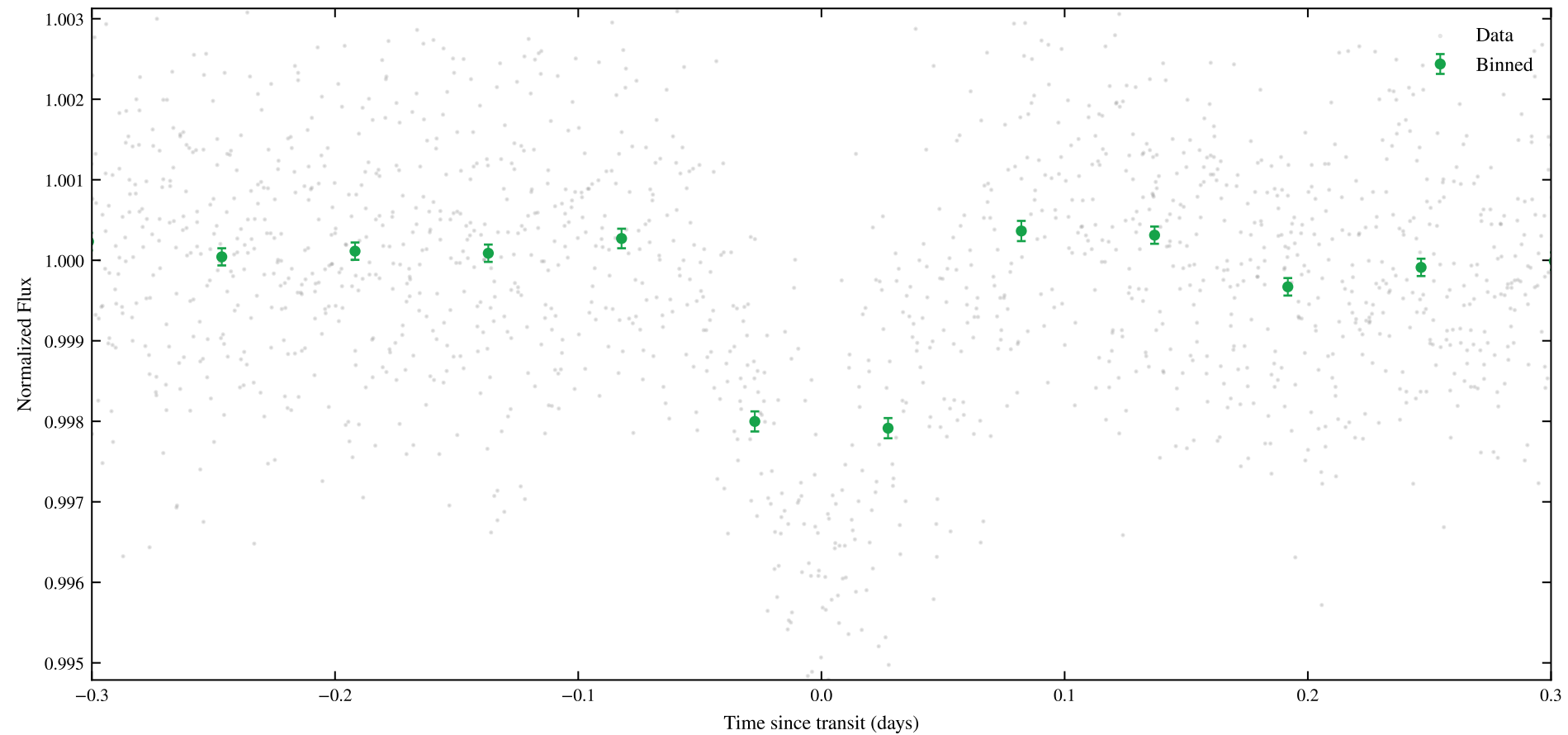
TIC 261369656 | Sectors: 104, 105 | TLS Coarse Periodogram (SDE = 8.34)



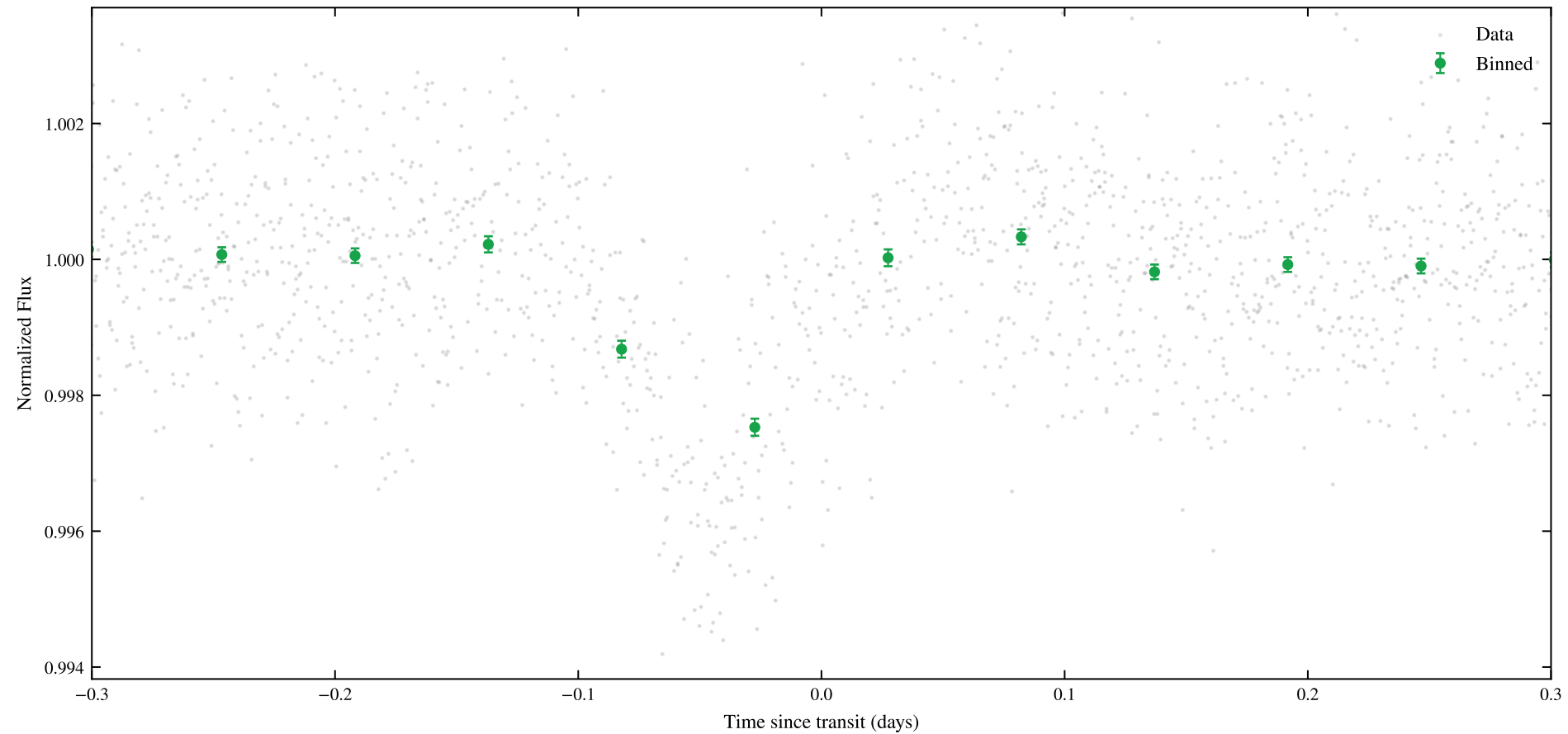
TIC 261369656 | Sectors: 104, 105 | TLS Fine Periodogram (SDE = 4.44)



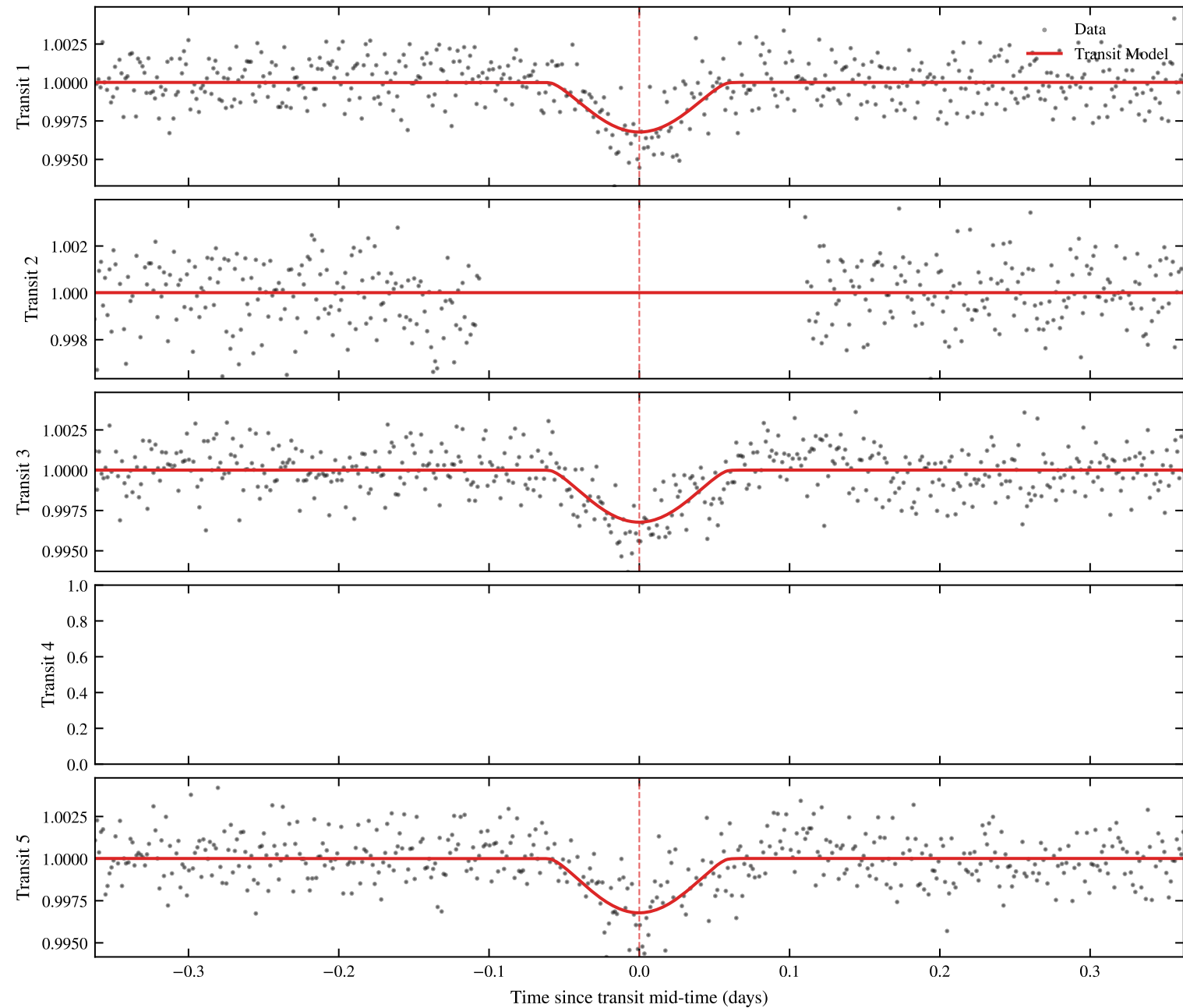
TIC 261369656 | Sectors: 104, 105 | Phase-Folded Transit (P = 10.95861 d)



TIC 261369656 | Sectors: 104, 105 | Phase-Folded Transit (P = 10.95815 d)



TIC 261369656 | Sectors: 104, 105 | Stacked Individual Transits



TIC 261369656 | Sectors: 104, 105 | Phased MCMC Fit - Planet 1

